

ABSTRACT

Weak passwords remain one of the most common cybersecurity vulnerabilities in higher education, threatening learning management systems, student portals, and university email services. This study investigates whether explainable machine learning models can detect weak passwords using only character-based password features. The Kaggle Password Strength Dataset was relabeled using zxcvbn (Wheeler, USENIX 2016), a real-world password strength estimator that captures dictionary words, keyboard walks, l33t-speak substitutions, and crack-time estimates — producing a more meaningful ground truth than simple composition rules. A balanced sample of 50,000 passwords was used, and nine row-local features were extracted. Logistic Regression, Decision Tree, and Random Forest classifiers were evaluated using stratified 5-fold cross-validation and a held-out test set, with scaling fitted inside a sklearn Pipeline to prevent data leakage. Random Forest achieved the best overall performance, while Decision Tree provided comparable results with transparent rule-based decisions. Explainable AI analysis showed that password length and entropy were the dominant indicators of password security, supporting both system-level defense and cybersecurity awareness in higher education environments.

RESEARCH QUESTION

How effectively can interpretable machine learning models detect weak passwords using character-based features, and how can these decisions support cybersecurity awareness in higher education environments?

BACKGROUND

Higher education institutions rely on online systems such as learning management platforms, student portals, university email services, and exam systems. Weak passwords expose students and staff to credential theft, unauthorized access, and account compromise. Explainable machine learning can identify risky password patterns and support cybersecurity awareness in academic environments.

OBJECTIVES

- Detect risky and secure passwords using machine learning
- Compare Logistic Regression, Decision Tree, and Random Forest models
- Apply explainable AI techniques to interpret model behavior
- Support password security awareness in higher education systems

EXPLAINABILITY AI (XAI) - FEATURE IMPORTANCE

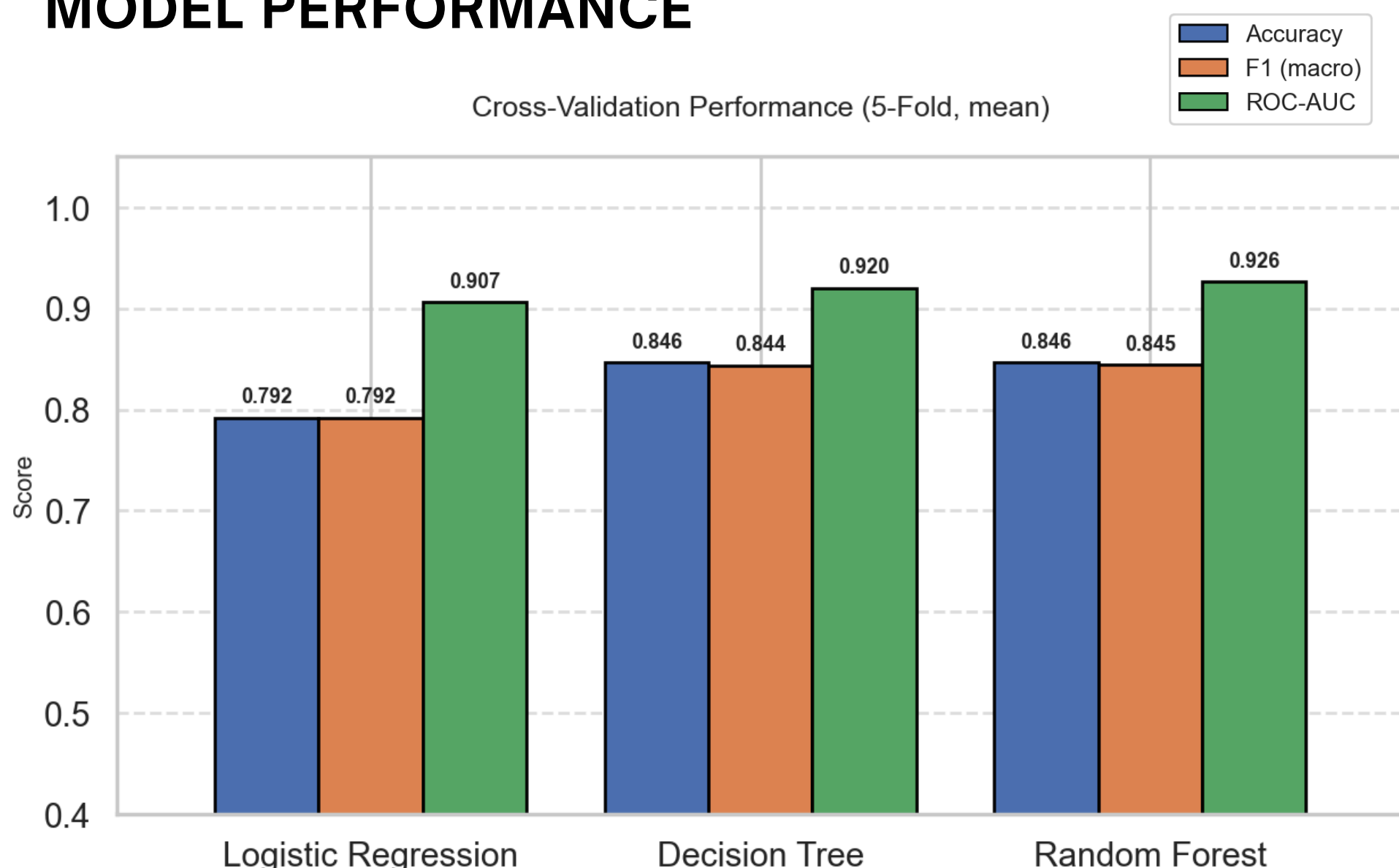
Random Forest achieved the best overall performance across all evaluation metrics, closely followed by Decision Tree, which produced comparable accuracy with more transparent decision rules.

Logistic Regression performed well and served as a stable, explainable baseline.

All three models converged on the same insight: password length and entropy are the dominant indicators of password strength, while short, sequential, or low-diversity passwords are strongly associated with weakness.

The remaining error gap reflects patterns that character-based features alone cannot capture — dictionary words, keyboard walks, l33t-speak substitutions, and date patterns — which require lexical and pattern-based analysis. This motivates hybrid password policies combining composition rules with breach-database lookups in higher education systems.

MODEL PERFORMANCE



CONCLUSION

Explainable machine learning successfully detected weak passwords using only character-based features. Random Forest achieved the strongest performance, while Decision Tree provided transparent, rule-based decisions.

Deploying interpretable models with combined with breach-database lookups can flag risky credentials at signup and support cybersecurity awareness in higher education environments.

DECISION TREE RULES

The Decision Tree translates password security into transparent rules, making the classification logic accessible for cybersecurity education.

Example rules:

- Passwords with length $\leq 8 \rightarrow$ almost always risky
- Medium-length passwords need higher entropy to be secure
- Long passwords with strong sequential patterns $\rightarrow$ risky
- Logistic Regression confirms entropy as the strongest secure signal and low diversity as the strongest risky signal

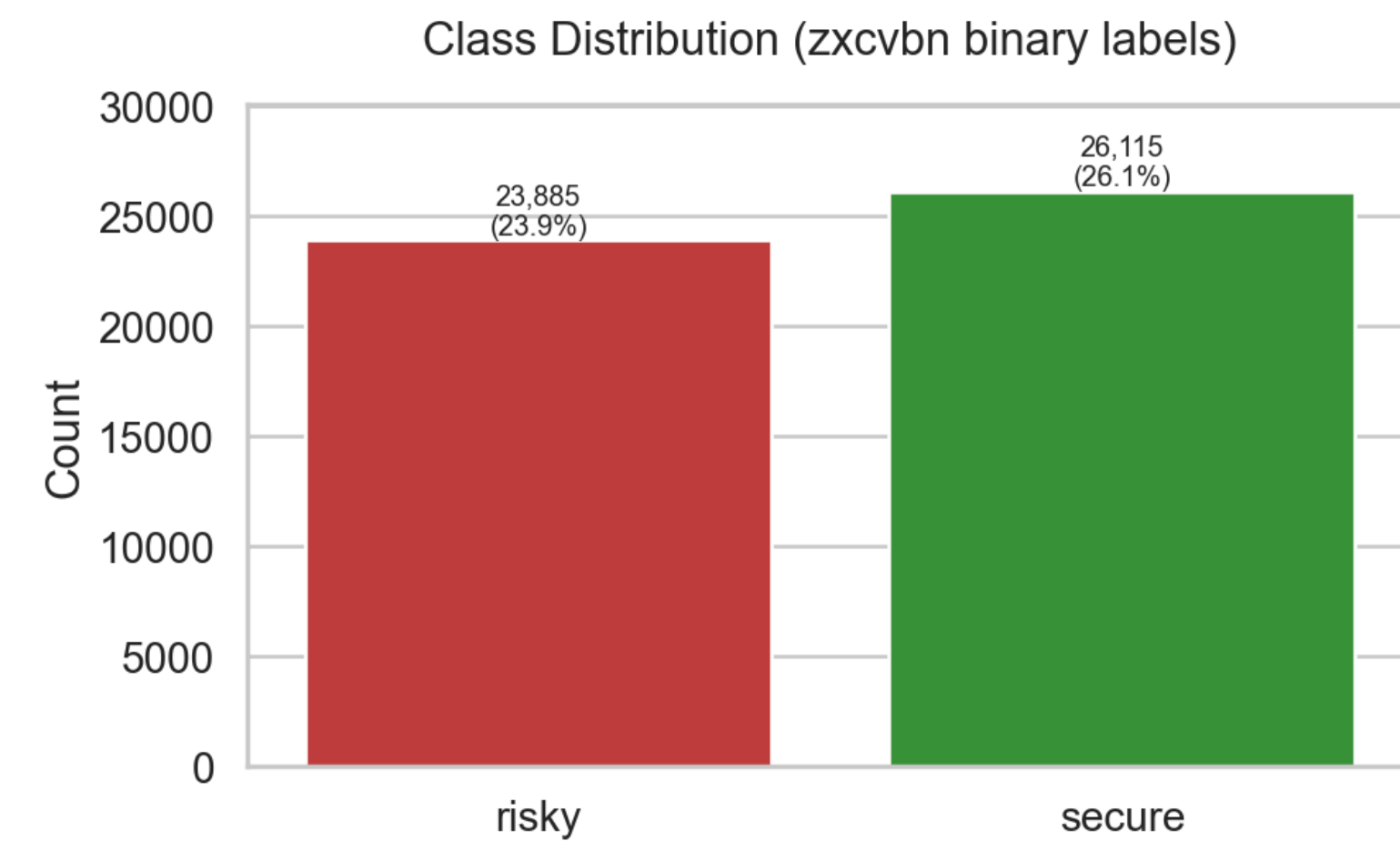
These rules make password classification interpretable and useful for cybersecurity awareness in higher education.

METHODOLOGY

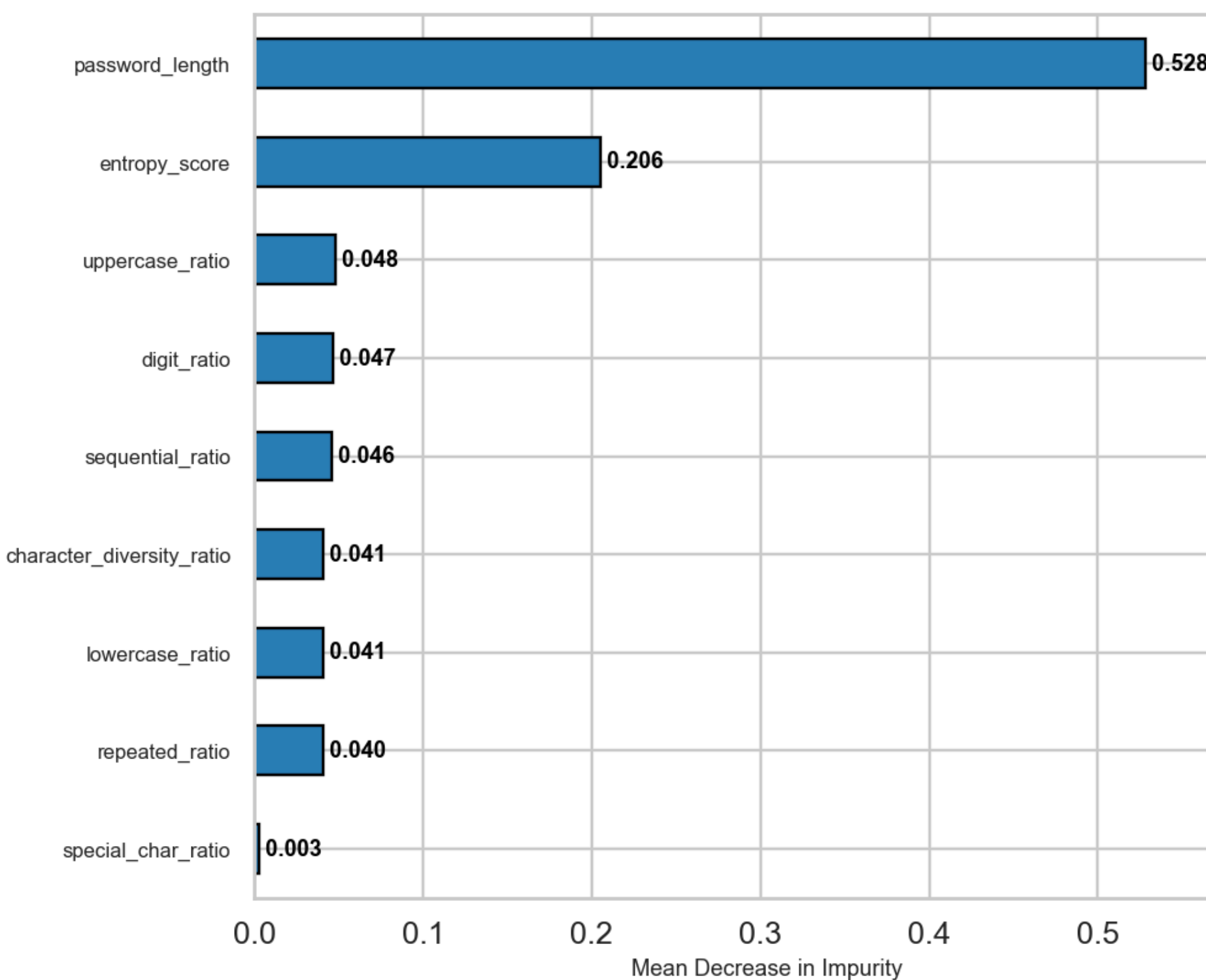
The Kaggle Password Strength Dataset (~670K rows) was relabeled using zxcvbn (Wheeler, USENIX 2016), which evaluates dictionary lookups, keyboard walks, l33t-speak substitutions, and crack-time estimation. Scores 0–2 were labeled as risky and 3–4 as secure. After cleaning, deduplication, and length filtering (1–72 characters), a random sample of 50,000 passwords was used, naturally producing an approximately balanced distribution. Nine row-local features — password length, Shannon entropy, character diversity, sequential and repeated patterns, and character class ratios — were extracted. Logistic Regression, Decision Tree, and Random Forest classifiers were evaluated using stratified 5-fold cross-validation. StandardScaler was fitted inside a sklearn Pipeline to prevent data leakage. RANDOM_STATE = 42 was used throughout for reproducibility.

DATASET SUMMARY

Item	Value
Cleaned dataset	669,634
Sampled passwords	50,000
Final features	9
Risky passwords	23,885
Secure passwords	26,115
Train / Test split	40K / 10K



FEATURE IMPORTANCE



Random Forest identified the most influential character-based features for password classification.

Most important features:

password_length, entropy_score, uppercase_ratio, digit_ratio, sequentialRatio

CONFUSION MATRIX RESULTS

Confusion matrices show how each model classifies risky and secure passwords on the held-out test set. Random Forest produced the most balanced classification, with low false-positive and false-negative rates across both classes. Decision Tree achieved similar performance with slightly higher false negatives. Logistic Regression showed more misclassifications in the risky class but remained reliable as a baseline.

